

Nosherwan Babar

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Professional Summary

MS Data Science candidate at Drexel University (GPA 3.75, graduating July 2026) with production experience building LLM-powered workflows, evidence extraction pipelines, and ML systems in Python. Built CyberSentinel: a production RAG pipeline over unstructured CVE data using LangGraph orchestration, LangChain, and ChromaDB – processing raw text into structured, actionable outputs. ML Engineer Co-op at URBN: built gradient-boosted models, Airflow ML lifecycle pipelines, drift detection, and model monitoring on GCP. Proficient with Claude Code and AI-powered developer tools as part of day-to-day development workflow.

Core Competencies

LLM-Powered Workflows & Evidence Extraction | LangGraph & LangChain Orchestration | Prompt Engineering & Structured Extraction | RAG Pipelines & Vector Databases | XGBoost & LightGBM (Tabular Classification) | Airflow (ML Lifecycle) | Model Monitoring & Drift Detection | Python | GCP | Docker | CI/CD

Work Experience

Machine Learning Engineer Co-op URBN (Urban Outfitters Inc.)

September 2025 – March 2026
Philadelphia, PA

- Built and deployed gradient-boosted and statistical regression models on tabular data with rigorous feature engineering, class-level evaluation, and offline experimentation before production rollout.
- Owned Airflow ML lifecycle pipelines on GCP: automated retraining workflows triggered by performance thresholds, drift detection systems monitoring production model health, and metrics/alerting for observability.
- Developed LLM-backed REST APIs using Python and FastAPI serving real-time predictions; integrated observability logging, error handling, and latency monitoring across the full inference stack.
- Implemented CI/CD pipelines with automated test coverage; wrote documentation and runbooks; participated in code reviews with cross-functional engineering and data science teams.

Junior Software Engineer Buggycy

August 2023 – August 2024
Lahore, Pakistan

- Built and maintained backend microservices and REST/gRPC APIs in Python and SQL; resolved query bottlenecks through optimization and caching, improving API latency by 15% on high-traffic endpoints.
- Wrote unit and integration test suites; configured CI pipeline for automated regression coverage; produced technical API documentation and developer guides for cross-team onboarding.

Projects

CyberSentinel – Production LLM-Powered Evidence Extraction Platform (Solo) *Python, LangGraph, LangChain, FastAPI, ChromaDB, GCP*

- Built an evidence extraction pipeline over live unstructured CVE databases: RAG pipeline with ChromaDB vector database and Gemini embeddings converts raw unstructured vulnerability text into structured, actionable outputs for downstream agent consumption.
- Designed 5-agent LangGraph orchestration with multi-step reasoning, tool-based execution, and prompt engineering with iterative tuning to improve retrieval accuracy and structured extraction quality.
- Deployed on GCP Cloud Run with GitHub Actions CI/CD, 332 passing tests, and LangChain tracing for end-to-end observability; integrated content safety guardrails throughout the pipeline.

Spending & Churn Analysis – Tabular Classification (Academic) *Python, XGBoost, LightGBM, PyTorch, Scikit-learn, PCA*

- Built end-to-end ML pipeline on multi-source transactional data: XGBoost and LightGBM gradient-boosted classifiers, PyTorch neural networks, PCA, and clustering; applied data balancing and outlier treatment to address skewed class distributions.
- Evaluated models with AUC, precision/recall, confidence intervals, and per-class performance breakdowns; ran offline experiments comparing model architectures before selecting production candidate.

Education

Master of Science in Data Science – Drexel University

GPA: 3.75/4.00 | NLP with Deep Learning, Applied ML, Recommender Systems, Quantitative Data Analysis

Graduating July 2026

Bachelor of Science in Computer Science – University of Central Punjab

GPA: 3.41/4.00 | Data Structures and Algorithms, Machine Learning, Statistics and Probability

July 2023

Skills

LLM/AI: LangGraph, LangChain, RAG, ChromaDB, Prompt Engineering, Structured Extraction, Gemini, OpenAI APIs, Claude Code

ML: XGBoost, LightGBM, PyTorch, Scikit-learn, Tabular Classification, Regression, Feature Engineering, Bias Analysis

MLOps: Airflow, Drift Detection, Model Monitoring, CI/CD (GitHub Actions), Automated Retraining, Experiment Tracking

Engineering: Python, FastAPI, REST/gRPC, Docker, SQL, Pandas, NumPy, Git, Automated Testing

Cloud: GCP (Cloud Run, BigQuery, Vertex AI), AWS S3, DBT, PySpark